

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - 5

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Class : SYIT	Batch : 1
Date of Assignment :	Date/Time of Submission :

Practical No. 5 - Write Algorithm, Code and Output

Q. 1. Write algorithm for the following:

a. Write a recursive function to print the factorial for a given number.

Code:

```
practical5a.py - C:/Users/mvluc/Desktop/laxmi phython/practical5a.py (3.14.6)
File Edit Format Run Options Window Help

def fact(n):
    if n==0:
        return 1
    else:
        return n * fact(n-1)

num=int(input("Enter your number:"))

print("The factorial of a number is", fact(num))
print("Laxmi jha S013")
|
```

Output:

```
>>> |=====
Enter your number:5
The factorial of a number is 120
Laxmi jha S013
>>> |
```

b. Write a function to check the input value is Armstrong or not.

Code:

```
pradeep@pradeep:~/Desktop/learn python/pradeep$ python3
File Edit Format Run Options Window Help
def armstrong(num):
    sum=0
    temp=num

    while(temp>0):
        digit=temp%10
        sum+=digit**3
        temp//=10

    if num==sum:
        print(num, "is a armstrong number")
    else:
        print(num, "is not a armstrong number")

num=int(input("Enter a number to check it is a armstrong number or not:"))
armstrong(num)
|
```

Output:

```
>>>
=====
Enter a number to check it is a armstrong number or not:554
554 is not a armstrong number
>>>
```

c. Write a function to check the input value is Palindrome or not .

Code:

```

practical5a.py - C:/Users/mvluc/Desktop/laxmi python/practical5a.py (3.14.6)
File Edit Format Run Options Window Help
def palindrome(num):
    n=num
    rev=0

    while(num!=0):
        rev=rev*10
        rev=rev+int(num%10)
        num=num//10

    if n==rev:
        print(n, "is a palindrome number")
    else:
        print(n, "is not a palindrome number")

num=int(input("Enter a number to check it is a palindrome number or not: "))

palindrome(num)
print("LAXmi Jha S013")

```

output:

```

>>> |===== RESTART:
Enter a number to check it is a palindrome number or not: 5
5 is a palindrome number
LAXmi Jha S013
>>> |

```

d. Write a lambda function that checks whether a given string starts with a specific character.

Code:

```

practical5a.py - C:/Users/mvluc/Desktop/laxmi python/practical5a.py
File Edit Format Run Options Window Help
starts_with_char = lambda s,c: s.startswith(c)

print(starts_with_char("hello", "h"))

print(starts_with_char("hello", "t"))

print(starts_with_char("Python", "p"))

print("LAXmi Jha S013")

```

output:

```
=====
True
False
False
LAXmi Jha S013
```

Q. 2. Write a python program to convert temperature from celcius to Fahrenheit. (Use the formula: $\text{fahrenheit} = (\text{celsius} * 9/5) + 32$)

Code: Using Ferahenitre

```
File Edit Format Run Options Window Help
celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9/5) + 32

print("Temperature in Fahrenheit:", fahrenheit)
```

Using Function:

```
practical5a.py - C:/Users/mvluc/Desktop/laxmi phython/practical5a.py (3.14.6)
File Edit Format Run Options Window Help
def fahrenheit(celsius):
    return (celsius * 9/5) + 32

celsius = float(input("Enter temperature in Celsius: "))

print("Temperature in Fahrenheit:", fahrenheit(celsius))
|
```

Output:

```
=====
Enter temperature in Celsius: 45
Temperature in Fahrenheit: 113.0
|
```

```
.>> Temperature in Fahrenheit: 113.0
=====
Enter temperature in Celsius: 32
Temperature in Fahrenheit: 89.6
.>> |
```

Q. 3. Write applications of use defined, built-in and external modules in